

Formulas

Regression:

$$\text{Slope} = \frac{r \times \text{SD}_Y}{\text{SD}_X}$$

$$\text{Intercept} = \text{Average}_Y - \text{Slope} \times \text{Average}_X.$$

Let A and B denote the names of any two events.

Multiplication Rule

The chance that both A and B happen equals the chance that A happens multiplied by the conditional chance that B happens *given* that A has happened.

Independence

A and B are independent if (and only if) the chance B happens given that A has happened is equal to the unconditional chance that B happens.

With Replacement

When drawing at random, *with* replacement, the *different* draws are independent.

Mutually Exclusive

A and B are mutually exclusive if they cannot both happen together.

Addition Rule

If A and B are mutually exclusive, then the chance that one happens is the *sum* of their chances.

Expanded Addition Rule

If A and B are *not* mutually exclusive, then the chance that *at least* one happens is the sum of their chances minus the chance that *both* happen.

Opposites Rule

The chance of an event plus the chance of its opposite is 1 (or 100%).

Expected Values and SEs

When drawing at random with replacement from a box of numbered tickets, the expected values and SEs of the sample sum, percentage, or average are:

Sample Estimate	Expected Value	Standard Error
Sum	Number of Draws $\times$ Average of Box	$\sqrt{\text{Number of Draws} \times \text{SD of Box}}$
Percent	Box Percentage	$\frac{\text{SE of Sum}}{\text{Number of Draws}} \times 100\%$
Average	Average of Box	$\frac{\text{SE of Sum}}{\text{Number of Draws}}$

SD of 2-value Box

If box (or sample) contains only two different values, the SD is given by:

$$\text{SD}_{\text{Box}} = (\text{Larger Value} - \text{Smaller Value}) \times \sqrt{(\text{Fraction of Larger}) \times (\text{Fraction of Smaller})}$$

Correction Factor

When sampling at random *without* replacement, the SEs above need to be multiplied by the finite correction factor, which is:

$$\sqrt{\frac{\text{Population Size} - \text{Sample Size}}{\text{Population Size} - 1}}$$

Confidence Intervals

The generic formula for a confidence interval is:

$$\text{Observed Sample Estimate} \pm (\text{Z-multiplier}) \times \text{SE of Sample Estimate}$$

where the multiplier is found from the Z-table for the required confidence level.